VECTORS

SCALAR QUANTITIES - have magnitude or size only.

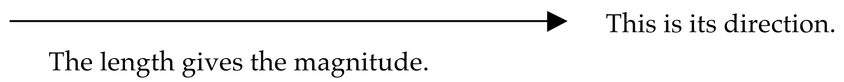
e.g. mass, distance, speed, density, temperature.

VECTOR QUANTITIES - have magnitude and direction.

e.g. displacement, velocity, acceleration, force, impulse, momentum.

Vectors are represented by arrows - the length indicates the magnitude of the vector and the arrowhead indicates the direction in which it acts.

i.e.



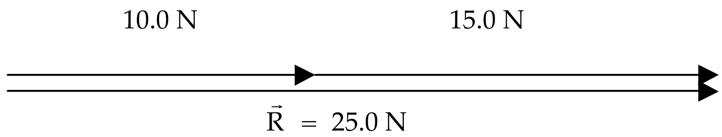
COMBINING VECTORS

ADDITION

Vectors are added head-to-tail. The resultant vector runs from the tail of the first vector to the head of the last vector.

1. **ADDITION**

ONE-DIMENSION

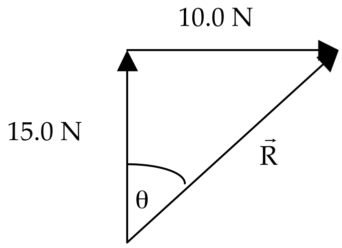


The resultant vector R = 25 .0 N - it runs from the tail of the first vector to the head of the last vector.

TWO DIMENSIONS

VECTORS AT RIGHT ANGLES

Use Pythagorus' Theorem and trigonometry to find the resultant.



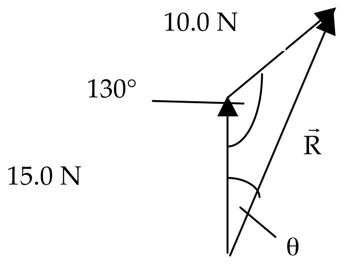
$$\vec{R} = \sqrt{(10.0)^2 + (15.0)^2} = 18.03 \text{ N}$$

$$\tan\theta = \frac{10.0}{15.0} \Rightarrow \theta = 33.69°$$

$\therefore$ <u>$\vec{R}$ = 18.0 N at 33.7° to the 15.0 N vector</u>.

VECTORS NOT AT RIGHT ANGLES

Use the sine and cosine rules for triangles to find the resultant vector.



$$\vec{R} = \sqrt{(15.0)^2 + (10.0)^2 - 2(15.0)(10.0)\cos 130^\circ}$$

$$= 22.76 \text{ N}$$

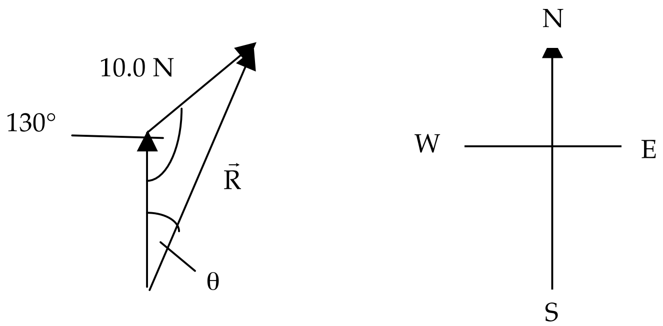
$$\frac{10.0}{\sin\theta} = \frac{22.76}{\sin 130^\circ}$$

$$\Rightarrow \theta = 19.67^\circ$$

$\therefore$ <u>$\vec{R}$ = 22.8 N at 19.7° to the 15.0 N vector</u>.

NOTE
If compass directions are given, it makes it easier to describe the direction.

Consider the example above.



The resultant vector R is at a 19.7° angle east of north.

$\therefore$ Direction is N 19.7° E.

Also, north can be considered the 0° bearing, with all other bearings determined clockwise from this direction.

$\therefore$ Direction is a bearing of 19.7°.

<u>Hence R = 22.8 N N 19.7° E or R = 22.8 N on a bearing of 19.7°</u>.

2. **SUBTRACTION**

ONE-DIMENSION

The easiest way to deal with vectors in one dimension is to **choose one direction as positive**.

Any vector in this direction is positive while all those in the other direction are negative.

It is then a simple matter of adding numbers.

EXAMPLE 1
A car moving at 15.0 ms^{-1} brakes suddenly to avoid a dog and reduces to a velocity of 5.00 ms^{-1}. Calculate its change in velocity.

Take the forwards direction as positive.

By definition the change in velocity is given by :

$$\Delta v = v - u$$

$$= 5.00\ ms^{-1} - 10.0\ ms^{-1}$$

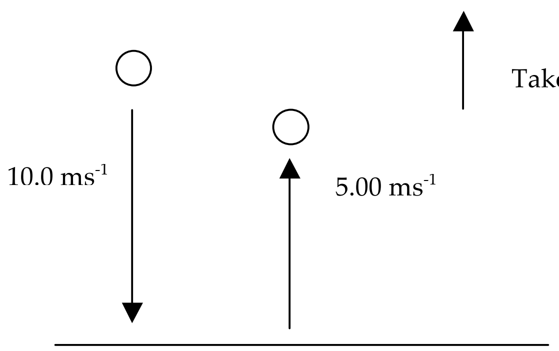
$$= -\ 5.00\ ms^{-1}$$

$\therefore$ <u>Change in velocity = 5.00 ms^{-1} backwards</u>.

The answer is negative which means the resultant is in the opposite direction to the one you chose as positive.

i.e. The direction is backwards.

EXAMPLE 2
A ball falls onto a concrete floor at 10.0 ms^{-1} and rebounds at 5.00 ms^{-1}. Calculate the change in velocity of the ball.



Take up as positive.

$u = -10.0\ ms^{-1}$ since it is down in the negative direction.

$$\Delta v = v - u$$

$$= 5.00 - (-10.0)$$

$$= \underline{15.0\ ms^{-1}\ \text{upwards}}$$

Answer is positive so the final direction is upwards in the positive direction chosen.

2. **TWO DIMENSIONS**

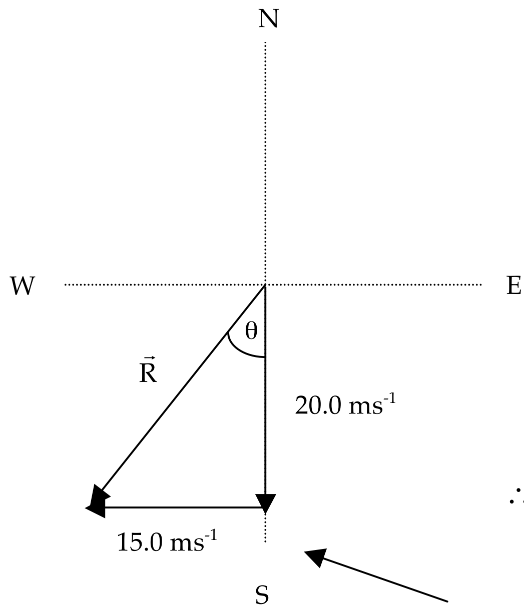
When subtracting vectors, the "negative" sign means:

"change the direction of the subtracted vector 180° and add it to the other vector".

This works best when vectors act at an angle to each other.

EXAMPLE

A car moving at 20.0 ms^{-1} north makes a left hand turn around a sweeping bend and heads west at 15.0 ms^{-1}. Calculate its change in velocity.



$$\Delta v = v - u$$

$$= 15.0\ ms^{-1}\ W - 20.0\ ms^{-1}\ N$$

$$= 15.0\ ms^{-1}\ W + 20.0\ ms^{-1}\ S$$

$$\vec{R} = \sqrt{(15.0)^2 + (20.0)^2} \qquad \tan\theta = \frac{15.0}{20.0}$$

$$= 25.0\ ms^{-1} \qquad \Rightarrow \quad \theta = 36.87°$$

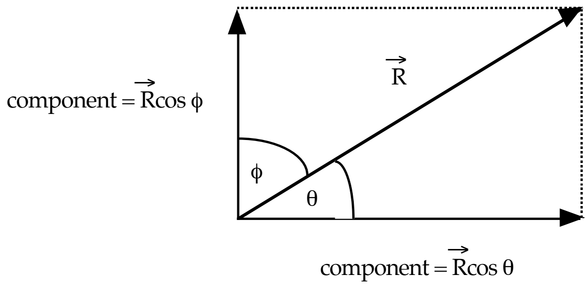
$$\therefore \quad \underline{\Delta v = 25.0\ ms^{-1}\ S\ 36.9°W}$$

The use of a compass grid line makes it easier to determine the direction of the resultant vector.

COMPONENTS OF A VECTOR

Any single vector can be expressed as ***two component vectors that are at right angles to each other***.

Neither component vector can be bigger than the original vector.



Any component is given by:

$$\boxed{\textbf{component} = \vec{\mathbf{R}}\cos\ \theta}$$

where R = given vector.
θ = the angle between R and the direction of the required component.

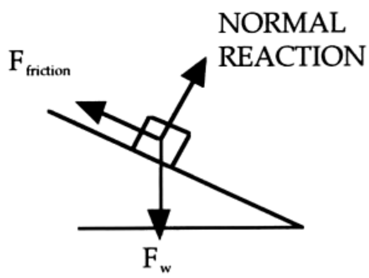
Examples

1. **Inclined Plane**

A box weighing 80.0 N sits at rest on a plane inclined at 27.0° to the horizontal.

(a) Draw a diagram showing all forces acting on the box.
(b) Calculate the force exerted perpendicular to the plane by the box onto the plane.

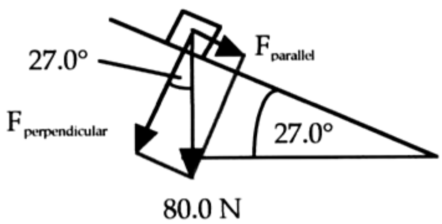
(a)



Only three forces act on the box.

The components of F_w are **not** forces that are acting.

(b)



$$
\begin{aligned}
F_{perp.} &= F_w \cos 27.0° \\
&= 80.0 \cos 27.0° \\
&= 71.28 \text{ N}
\end{aligned}
$$

$\therefore$ $F_{perp.}$ = 71.3 N into the plane.

2.

A box of mass 12.5 kg is sliding down a frictionless plane, inclined at 35° to the horizontal. Determine the force.

(a) (i) acting down the plane
(ii) into the plane

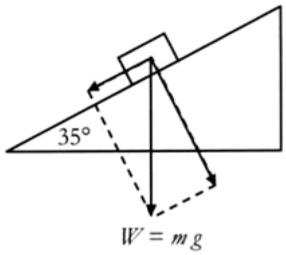
(b) Assuming $g = 9.80\ ms^{-2}$, determine the acceleration of the mass down the plane.

(a) Force || to the plane

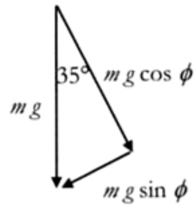
$$F_{\parallel} = m\,g \sin\theta$$
$$= (12.5)(9.80)(\sin 35°)$$
$$= 70.3\ N$$

Force perpendicular to the plane

$$F_{perp} = m\,g \cos\theta$$
$$= (12.5)(9.80)(\cos 35°)$$
$$= 100\ N$$



or



(b) Since $F = m\,a$

$$a = \frac{F}{m}$$

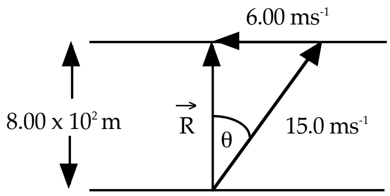
|| to plane $a = \dfrac{mg \sin\theta}{m}$

$$a = (9.8)\ (\sin 35)$$
$$= 5.62\ ms^{-2}$$

3. **River Problem**

A person wants to cross a river 8.00×10^2 m wide to reach a jetty directly across on the opposite bank. If the river is flowing at $6.00\ ms^{-1}$ and the boat that is used can travel at $15.0\ ms^{-1}$ in still water, calculate:

(a) the resultant velocity of the boat in relation to the river bottom (or bank).
(b) the time taken to reach the other side.



(a) $\vec{R} = \sqrt{(15.0)^2 - (6.00)^2}$
$= 13.75\ ms^{-1}$

$\therefore$ $\underline{\vec{R} = 13.8\ ms^{-1} \text{ at } 90.0° \text{ to the bank.}}$

(b) To calculate the time taken, use a component of any velocity that is directly across the river.

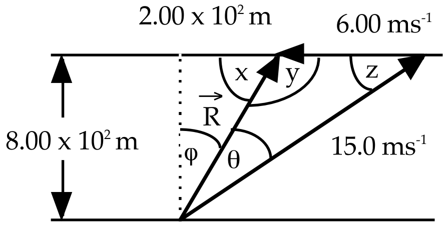
In this example, $\vec{R}$ is directly across so use that.

i.e. $v_{across} = \dfrac{s_{across}}{t_{across}}$

=> $t = s\ /\ v$
$= 8.00 \times 10^2\ /\ 13.75$
$= 58.18\ s$

$\therefore$ Time taken = 58.2 s .

4. If the person in Question 3 was trying to reach a jetty 2.00×10^2 m upstream, calculate the resultant velocity and the time taken.



$\tan \phi = 2.00 \times 10^2 \ / \ 8.00 \times 10^2$

$\Rightarrow \phi = 14.04°$

By geometry: $x = 75.96°$

$y = 104.04°$

To find θ: $\dfrac{6.00}{\sin \theta} = \dfrac{15.0}{\sin 104.04°}$

$\Rightarrow \quad \theta = 22.83°$

By geometry: $z = 53.13°$

To find R: $\dfrac{\vec{R}}{\sin 53.13°} = \dfrac{6.00}{\sin 22.83°}$

$\Rightarrow \quad \vec{R} = 12.37 \text{ ms}^{-1}$

$\therefore$ $\underline{\vec{R} = 12.4 \text{ ms}^{-1} \text{ at } 53.1° \text{ to the bank upstream.}}$

To calculate the time taken to cross, use any component of velocity that is directly across the river.

e.g. $12.37 \cos 14.04°$ or $15.0 \cos 36.87°$

$$v_{across} = \frac{s_{across}}{t_{across}}$$

$$\begin{aligned} \Rightarrow \quad t &= s \ / \ v \\ &= 8.00 \times 10^2 \ / \ 12.37 \cos 14.04° \\ &= 66.66 \text{ s} \end{aligned}$$

$\therefore$ Time taken = 66.7 s .